INSTITUTE OF COMPUTER AND INFORMATION SCIENCES

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BANKURA UNIVERSITY

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DECLARATION

I, **Meghna Das**, hereby declare that the project titled **"Analysis of Diseases Caused Due to Global Warming in Different States of India Using Power BI"** is an original work carried out by me under the guidance of my mentor **Mr. Biswarup Mandal**.

I further declare that this project has been completed as part of the academic requirements of my BBA program and has not been submitted earlier, either in part or in full, to any university or institution for the award of any degree, diploma, or certificate.

All the information, data, and analysis presented in this project are based on secondary data collected from reliable sources. Proper acknowledgements have been made wherever the work of others has been referred to.

I take full responsibility for the authenticity and originality of this work.

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Date: ___17.12.2025____

Industrial Internship Certificate

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This certificate is awarded to

MEGHNA DAS

of

INSTITUTE OF COMPUTER AND INFORMATION SCIENCES

for successfully completing the **Industrial Internship** on

POWER BI

from

SEPTEMBER 12, 2025 to DECEMBER 12, 2025

and implementing the project titled

CLIMATE CHANGE AND DISEASE IN INDIAN STATES USING POWER BI

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EXECUTIVE SUMMARY

Global warming has emerged as a critical global issue with serious implications for human health. Rising temperatures, changing rainfall patterns, increased humidity, and deteriorating air quality have significantly contributed to the spread and intensity of climate-sensitive diseases. India, due to its diverse climatic conditions and large population, is highly vulnerable to these health impacts. This project aims to analyse the diseases caused due to global warming in different states of India using **Power BI** as a data analytics and visualization tool.

The study is based on secondary data collected from government health reports, climate databases, and published research sources. Climate variables such as temperature, rainfall, humidity, and air quality were analysed alongside disease data related to vector-borne, water-borne, respiratory, and heat-related illnesses. A quantitative and analytical research approach was adopted to examine state-wise and seasonal variations in disease patterns.

The Power BI dashboards developed in this project provide a clear and interactive representation of climate–health relationships. The analysis reveals that vector-borne diseases such as dengue and malaria account for the highest number of cases, particularly in states with high humidity and heavy rainfall. Respiratory illnesses were found to cause the highest number of deaths, closely associated with poor air quality and rising temperatures. Heat-related illnesses showed a significant increase during extreme temperature periods, especially in heat-prone states.

The study also highlights considerable state-wise and rural–urban disparities. Rural areas showed higher vulnerability due to limited healthcare infrastructure, while seasonal analysis confirmed sharp increases in disease cases during monsoon and summer months. These findings underline the need for region-specific climate and health strategies.

Overall, the project demonstrates the effectiveness of **Power BI** in transforming complex climate and health data into meaningful insights. The study emphasizes the importance of data-driven decision-making in addressing climate-induced health risks and supports the development of preventive public health strategies and climate-resilient healthcare systems in India.

LITERATURE REVIEW

Climate change and global warming have become central themes in modern scientific and public health research. Over the past two decades, scholars, environmental agencies, and health organizations have increasingly investigated the relationship between rising temperatures, changing rainfall patterns, humidity levels, and the emergence of new health risks. This literature review brings together key findings from national and international studies, focusing on how climate variations influence the occurrence and spread of diseases, particularly in the context of India's diverse climatic conditions.

1. Global Research on Climate Change and Health:

Early research by the Intergovernmental Panel on Climate Change (IPCC) highlighted that global warming significantly increases the vulnerability of populations to vector-borne, waterborne, and heat-related diseases. IPCC's Assessment Reports emphasize that warmer temperatures accelerate the life cycle of disease vectors such as mosquitoes, allowing species like *Aedes aegypti* and *Anopheles* to thrive in new environments. Studies published in *The Lancet Planetary Health* confirm that climate-sensitive diseases such as dengue, malaria, chikungunya, cholera, and diarrheal illnesses are being reported in regions where they were previously rare.

Furthermore, research from the World Health Organization (WHO) connects extreme heat events with increased mortality from dehydration, heatstroke, and cardiovascular stress. WHO also reports that air pollution, when combined with climate change, contributes to respiratory problems, asthma attacks, and chronic lung diseases. These findings collectively establish that global warming represents a major public health threat, influencing almost every continent.

2. Climate Change and Disease Patterns in India:

Indian researchers and national bodies have also contributed significantly to understanding climate-health linkages. According to studies by the Indian Council of Medical Research (ICMR), India's climatic diversity makes it highly susceptible to climate-sensitive diseases. Research indicates that vector-borne diseases such as dengue and malaria rise during periods of high humidity and temperature, particularly in states like Kerala, Karnataka, West Bengal, and Maharashtra. Studies conducted in the northeastern states show that prolonged monsoon seasons and increasing rainfall intensity create favorable breeding conditions for mosquitoes, leading to spikes in malaria and Japanese encephalitis.

Public Health Foundation of India (PHFI) reports emphasize the increased incidence of waterborne diseases such as cholera, typhoid, and diarrhea during excessive rainfall or flooding. These diseases are particularly common in low-lying states such as Assam, Bihar, and Odisha, where drainage and sanitation systems are frequently overwhelmed by heavy rain.

Similarly, research from the Centre for Science and Environment (CSE) highlights the connection between rising temperature levels and respiratory diseases in metropolitan cities like Delhi, Mumbai, and Kolkata. Higher temperatures combined with vehicular pollution intensify ground-level ozone formation, which directly affects lung health. These studies collectively show that the health impacts of climate change vary significantly from state to state due to India's diverse environmental and socio-economic conditions.

3. Data Analytics in Climate and Health Research:

With the increasing availability of large datasets, researchers have begun using data analytics, forecasting tools, and visualization platforms to study climate-related health trends. Scientific papers from the field of environmental informatics emphasize the role of data analytics in creating predictive models that can estimate future disease outbreaks based on climate indicators such as temperature, humidity, and rainfall.

Tools like Power BI, R, and Python have been widely adopted to analyze climate-health relationships. Researchers argue that data visualization tools allow policymakers and health officials to quickly interpret complex datasets and identify patterns that are not visible using traditional analysis. A growing number of studies show that interactive dashboards and heatmaps provide better decision-support during public health emergencies.

4. Literature Gaps and Need for the Present Study:

Although various studies have explored climate change and health separately, limited research exists that systematically compares **state-wise disease patterns in India using data analytics tools like Power BI**. Most existing literature focuses either on specific diseases or on regional climatic changes, but few studies integrate **climate variables + health data + visual analytics** to produce a holistic, state-level comparison.

There is also a gap in literature regarding **real-time data visualization**, which is crucial for understanding rapidly changing climate conditions. The need for interdisciplinary tools to simplify complex datasets is repeatedly highlighted by researchers, suggesting that integrating climate science, public health, and analytics is essential for future disease prediction and prevention.

This project addresses these gaps by:

- Combining climate data and disease data
- Applying modern analytical tools
- Performing state-wise comparisons across India
- Creating interactive dashboards for visual interpretation

Thus, this literature review establishes the foundation for a more advanced, data-driven exploration of climate-induced diseases in India.

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SYNOPSIS

Title of the Project

Analysis of Diseases Caused Due to Global Warming in Different States of India Using Power BI

➢ Introduction

Global warming has become a major global concern, significantly affecting environmental stability and human health. Rising temperatures, irregular rainfall, increased humidity, and deteriorating air quality have contributed to the spread of climate-sensitive diseases. India, due to its diverse climate and large population, is highly vulnerable to these changes. Different states experience varying disease patterns depending on climatic conditions. This project aims to analyse the impact of global warming on disease occurrence across Indian states using data analytics.

➢ Objectives of the Study

1. To examine the impact of global warming on disease patterns in India.
2. To analyse state-wise distribution of climate-induced diseases.
3. To study the relationship between climate factors and disease occurrence.
4. To identify high-risk states and vulnerable populations.
5. To use Power BI for effective data visualization and analysis.

Scope of the Study

The study focuses on selected climate-sensitive diseases across different states of India. It is based on secondary data collected from government and public health sources. The analysis is limited to state-level data and does not include individual clinical records.

➢ Research Methodology

A quantitative and analytical research design has been adopted. Secondary data related to climate variables and disease incidence were collected, cleaned, and analysed using Power BI. The study involves trend analysis, state-wise comparison, and visualization through interactive dashboards.

➢ Tools Used

- Power BI
- Microsoft Excel
- Secondary data sources (Government reports, WHO, ICMR, IPCC)

- **Significance of the Study**

The study highlights the growing health risks associated with climate change in India and demonstrates the importance of data-driven analysis for public health planning. It also showcases the practical application of Power BI in analysing real-world climate and health data.

- **Expected Outcome**

The project is expected to identify climate-sensitive diseases, highlight high-risk regions, and provide clear visual insights that can support healthcare planning and policy decision-making.

- ✓ **Name of the Student:** Meghna Das
- ✓ **Project Guide:** Mr. Biswarup Mandal

INTRODUCTION

Global warming has become one of the most pressing challenges of the modern world, influencing not only the environment but also human health, economies, and the overall functioning of societies. Over the past few decades, the rise in Earth's average temperature has intensified climate variability, leading to more frequent heatwaves, irregular monsoon cycles, increasing humidity levels, severe floods, cyclones, and extended droughts. These climatic shifts are no longer distant environmental concerns; they now represent a significant **public health emergency**. As global warming accelerates, the interconnection between climate change and human health is becoming more evident, creating new disease patterns and worsening existing ones.

Global Warming as a Growing Health Threat: The scientific community has consistently warned that rising temperatures create ideal conditions for various pathogens and disease vectors to thrive. Warmer climates allow mosquitoes, bacteria, viruses, fungi, and parasites to expand into new geographical areas. Increased humidity supports the growth of waterborne pathogens, while unpredictable rainfall patterns contaminate drinking water sources. Heatwaves, air pollution, forest fires, and degraded ecosystems contribute to respiratory illnesses, cardiovascular problems, and heat-related disorders. This means global warming, directly and indirectly, affects human health in multiple ways, making it one of the fastest-growing health threats across the globe.

According to global research reports, climate-linked diseases have risen sharply in the last two decades. Many diseases that were previously limited to certain regions are now emerging in new states and countries due to shifting temperatures. Heat strokes, dehydration, asthma, malnutrition, vector-borne diseases (like malaria and dengue), cholera, typhoid, leptospirosis, skin infections, and zoonotic outbreaks are directly linked with climate variations. These trends show that understanding climate-related diseases is no longer optional—it is **essential for public health planning, policy formulation, and disaster management**.

India as One of the Most Climate-Vulnerable Countries: India is identified as one of the most climate-vulnerable nations in the world due to its diverse geography, large population, dependence on monsoon cycles, and economic exposure to climate-sensitive sectors. With more than 1.4 billion people spread across coastal regions, mountains, plains, deserts, agricultural lands, and densely populated cities, India experiences climate change impacts more intensely and unevenly across states.

The country faces:

- **Extreme heatwaves** in Rajasthan, Delhi, Uttar Pradesh, and Telangana
- **Heavy rainfall and flooding** in Mumbai, Chennai, Assam, Odisha, and Kerala
- **Cyclones** in coastal states like West Bengal, Odisha, Tamil Nadu, and Andhra Pradesh
- **Rising vector-borne diseases** in West Bengal, Kerala, Karnataka, Maharashtra, and Gujarat
- **Waterborne diseases** in Bihar, Assam, Jharkhand, and Uttar Pradesh
- **Severe air pollution** in Delhi NCR, Punjab, Haryana, and Maharashtra

- **Zoonotic outbreaks** in southern and northeastern states
- **High humidity–related fungal diseases** in coastal regions

Each state experiences climate change differently, and therefore, the disease patterns also vary. For example:

- **Dengue, malaria, and chikungunya** rapidly grow in warm and humid climates.
- **Heatstroke and dehydration** rise in extremely hot states.
- **Cholera and diarrheal diseases** increase during excessive rainfall or waterlogging.
- **Respiratory diseases** worsen in polluted and high-temperature regions.

These patterns clearly show that global warming amplifies disease risks across India, making **state-wise disease analysis a crucial step** towards understanding national health vulnerability.

Increase in Climate-Sensitive Diseases Across India: Data collected over the last 10–15 years shows that climate change has led to a noticeable increase in:

1. **Vector-Borne Diseases:** Warmer climates accelerate mosquito breeding cycles and expand their habitat. Diseases like dengue, malaria, chikungunya, and Japanese encephalitis have significantly increased in states like West Bengal, Assam, Kerala, Uttar Pradesh, and Maharashtra.
2. **Waterborne Diseases:** Irregular rainfall, floods, and water contamination contribute to cholera, typhoid, diarrhea, and hepatitis outbreaks. These diseases spike especially after monsoon seasons.
3. **Heat-Related Illnesses:** States experiencing heatwaves—Rajasthan, Delhi, Punjab, Haryana, Telangana—report high cases of heat exhaustion, sunstroke, kidney stress, and dehydration.
4. **Respiratory Diseases:** Air pollution combined with higher temperatures leads to asthma, bronchitis, COPD, and allergies. Pollution combined with heat increases ground-level ozone, which is harmful for lungs.
5. **Skin and Fungal Diseases:** High humidity and warm temperatures support fungal growth, causing frequent skin issues in coastal states like West Bengal, Odisha, Kerala, and Tamil Nadu.
6. **Zoonotic Diseases:** Deforestation and habitat loss cause closer human–animal interaction, leading to diseases like Nipah, H1N1, bird flu, and swine flu.

These diseases highlight the urgent need for a **data-driven approach** to identify trends, predict risks, and prepare healthcare strategies.

Importance of a Data-Driven Analysis Using Power BI: Traditional health data reports are often static, unstructured, and difficult to interpret. For a complex topic like climate–disease interaction, advanced analytics is essential. Power BI provides the perfect platform for integrating climate, geography, and health datasets, helping researchers analyze patterns with accuracy and clarity.

Power BI benefits for this project:

Data Integration: Allows combining multiple datasets such as temperature, rainfall, humidity, disease counts, and state-wise population.

Interactive Visualisation: Heatmaps, charts, graphs, and maps make the analysis easy to understand for both academic and professional audiences.

State-Wise Disease Insights: Comparison of disease growth across all Indian states.

Trend Identification: Helps identify how diseases rise or fall with climate factors.

Forecasting & Predictive Analysis: Power BI's built-in AI visuals and forecasting help predict future risks and seasonal patterns.

Professional Reporting: Creates clean, modern dashboards used in real-world industries, healthcare, disaster management, and government policymaking.

This project transforms raw data into **meaningful insights**, making it suitable for research, academic submissions, and documentary creation.

Purpose of Combining Climate Science + Health Data + Data Analytics

This project serves a multidisciplinary purpose:

- To understand the real impact of global warming on human health
- To analyze how different states of India react differently to climate variations
- To use real data to identify the diseases most affected by temperature, humidity, and rainfall
- To provide a powerful visual representation using Power BI dashboards
- To support health planning, climate adaptation, and future policy recommendations
- To create a strong research contribution for your BBA project and documentary

This combination makes your report **scientifically strong and analytically impactful**.

✓ **Why Power BI Was Chosen for This Project:** Power BI was selected because:

- It supports **large datasets** without slowing down
- It offers **state-wise India maps**, helping in geographical disease analysis
- It gives **clear visual insights** suitable for documentaries
- It helps identify **hidden patterns** between climate factors and diseases
- It is widely used in corporate organizations, healthcare analytics, and environmental research

Thus, Power BI strengthens both the academic and professional value of the project.

ABOUT POWER BI

Power BI is a **business intelligence (BI) and data visualization tool** developed by **Microsoft**. It is used to **collect, analyze, and visualize data** in an easy and interactive way.

What Power BI Does

- Converts **raw data** into **meaningful insights**
- Creates **interactive dashboards and reports**
- Helps in **better decision-making** using data

Main Components of Power BI

1. **Power BI Desktop** – Used to create reports and dashboards
2. **Power BI Service** – Cloud platform to publish and share reports
3. **Power BI Mobile App** – View reports on mobile devices

Key Features

- Supports data from **Excel, SQL, CSV, Web, APIs**, etc.
- **Drag-and-drop** visuals (charts, maps, tables)
- Real-time data updates
- Filters, slicers, and drill-down options
- Secure data sharing and collaboration

Uses of Power BI

- Business performance analysis
- Sales and marketing reports
- Financial analysis
- Academic and research projects (like your **Power BI internship project**)

Advantages

- User-friendly interface
- Fast data processing
- Professional-looking dashboards
- Integrates well with Microsoft tools

Conclusion

Power BI is a powerful tool that helps organizations and students **analyze data efficiently**, **identify trends**, and **make informed decisions** using visual reports.

OBJECTIVES OF THE PROJECT

The primary objective of this project is to analyze the impact of global warming on the spread and intensity of diseases across different states of India using modern data analytics tools. The following specific objectives guide the study:

1. To identify the major diseases influenced by climate change in India.

This includes examining vector-borne, waterborne, heat-related, respiratory, skin, and zoonotic diseases that are directly or indirectly linked to temperature, humidity, and rainfall variations.

2. To analyze state-wise climate variations and their effect on disease patterns.

The project seeks to compare how different states experience varying climatic conditions—such as heatwaves, heavy rainfall, humidity, and floods—and how these environmental shifts influence disease prevalence.

3. To study the relationship between climate variables and health outcomes.

A key objective is to establish correlations between climate indicators (temperature, rainfall, humidity) and disease outbreaks using quantitative data.

4. To integrate climate data and health data into a single analytical framework.

The project aims to combine datasets from different sources to create a unified model that supports deeper understanding of climate-induced health risks.

5. To use Power BI for visualizing disease trends and climate impacts.

Another important objective is to design interactive dashboards, maps, graphs, and charts that present complex climate–disease relationships in a clear and understandable manner.

6. To identify high-risk states based on climate sensitivity and disease burden.

This helps in understanding which regions require urgent health interventions or climate adaptation measures.

7. To support evidence-based decision-making for public health and climate policy.

The study aims to provide insights that can assist policymakers, healthcare systems, researchers, and environmental planners in developing strategies to mitigate climate-related health threats.

8. To evaluate how global warming may influence future disease trends.

Through trend analysis and visual forecasting tools, the project aims to predict potential disease risks that may emerge as climate conditions continue to evolve.

9. To demonstrate the practical application of data analytics in solving real-world problems.

This objective highlights the importance of tools like Power BI in transforming raw environmental and health data into meaningful, actionable insights.

10. To reflect how Ardent Computech training contributed to building industry-relevant analytical skills.

The project also aims to showcase how hands-on training in Power BI enabled the creation of professional datasets, dashboards, and reports.

SCOPE OF THE STUDY

The scope of this project, *"Analysis of Diseases Caused Due to Global Warming in Different States of India Using Power BI,"* covers a comprehensive examination of climate-linked health impacts across India. The study focuses on understanding how rising temperatures, irregular rainfall, extreme weather events, and other climate-related changes contribute to variations in disease patterns at the state level. This section outlines the boundaries, areas of coverage, and the overall extent of the research.

1. Geographical Scope:

- The study includes **all major states and union territories of India**, representing diverse climates such as tropical, desert, coastal, Himalayan, and plateau regions.
- Disease patterns are compared across:
  - **Coastal regions** (West Bengal, Odisha, Kerala, Andhra Pradesh, Tamil Nadu)
  - **Northern plains** (Uttar Pradesh, Haryana, Punjab, Bihar)
  - **Western and central regions** (Rajasthan, Gujarat, Maharashtra, Madhya Pradesh)
  - **North-eastern states** (Assam, Meghalaya, Manipur, Mizoram)
  - **Himalayan states** (Himachal Pradesh, Uttarakhand, Jammu & Kashmir)
- This wide coverage helps in identifying **region-specific vulnerabilities** and climate–disease relationships.

2. Disease Scope:

The study focuses on diseases that are scientifically linked to climate change, especially:

- **Vector-Borne Diseases**
  - Malaria
  - Dengue
  - Chikungunya
  - Zika
  - Japanese Encephalitis
- **Water-Borne Diseases**
  - Cholera
  - Diarrhoea
  - Typhoid
  - Hepatitis A
- **Heat-Related Diseases**
  - Heat stroke
  - Dehydration
  - Cardiovascular stress
- **Air-Related and Pollution-Linked Diseases**
  - Asthma
  - Respiratory infections
- **Zoonotic Diseases (Climate-Sensitive)**
  - Leptospirosis

- Nipah
- Avian influenza

By focusing on these categories, the study captures a **wide spectrum of global-warming-sensitive diseases** that affect different states differently.

3. Climate Variables Covered:

The analysis examines the influence of key climatic factors, including:

- **Temperature rise** (annual & seasonal)
- **Rainfall patterns** (floods, heavy rainfall, monsoon shifts)
- **Humidity levels**
- **Extreme weather events** (cyclones, droughts, heatwaves)
- **Water stagnation and flooding intensity**

These variables are crucial to understanding the **environmental triggers** behind disease outbreaks.

4. Time Scope:

- The study considers data from **recent years**, preferably covering a **5–10 year period**, depending on availability of datasets.
- Long-term patterns help identify:
  - disease seasonality,
  - rising trends,
  - climate-linked outbreak cycles.

5. Data Scope:

- The project uses **publicly available and secondary data**, such as:
  - National Health Mission (NHM) reports
  - Ministry of Health & Family Welfare datasets
  - National Centre for Vector Borne Disease Control (NCVBDC)
  - IMD (Indian Meteorological Department) climate data
  - State health department records
- All data is cleaned, organized, and analysed using **Power BI visual tools**.

6. Analytical Scope:

Using Power BI, the study covers:

- State-wise comparison of climate-sensitive diseases
- Trend analysis of disease cases over time
- Visualization of disease hotspots
- Correlation between temperature/rainfall and disease rise
- Seasonal patterns of vector and water-borne diseases
- Identification of high-risk states and disease clusters

Power BI dashboards provide clear, interactive insights that help in drawing meaningful conclusions.

7. Functional Scope:

The study supports:

- **Public health planning**
- **Early warning systems** for outbreaks
- **Identification of vulnerable states and populations**
- **Academic research** on climate-health interactions
- **Data-driven decision-making** by policymakers

8. Limitations (Included as part of Scope Definition):

Though the study is extensive, it acknowledges limitations:

- Relies on secondary data availability
- Some states may have incomplete reporting
- Power BI visualizations depend on quality and completeness of datasets
- Does not include hospital-level clinical analysis

➢ Overall, Scope Statement:

This project provides a **state-wise, disease-wise, and climate-variable–wise analysis** of how global warming influences disease outbreaks in India. It focuses on understanding patterns, identifying vulnerable regions, and visually representing findings through Power BI dashboards to support research, planning, and policy development.

METHODOLOGY

The methodology outlines the systematic steps used to conduct the study *"Analysis of Diseases Caused Due to Global Warming in Different States of India Using Power BI."*
This section explains the **research design**, **data sources**, **variables**, **tools used**, **data preparation steps**, and **analytical techniques** used in the project.

1. Research Design:

- ✓ **Type of Research:** This study is based on:
  - **Descriptive Research** – to describe patterns of climate-sensitive diseases.
  - **Analytical Research** – to examine relationships between climate variables and disease incidence.
  - **Comparative Research** – to compare disease data across different states of India.
- ✓ **Approach Used:** A **quantitative, data-driven approach** is used because the study involves:
  - numerical datasets
  - climate measurements
  - disease incidence data
  - statistical and visual analysis using Power BI

2. Data Collection Method:

- ✓ **Nature of Data:** The study uses **secondary data**, which means data collected from already published government and research sources.
- ✓ **Data Sources:** Data is collected from:
  - **National Health Mission (NHM)** – disease incidence data
  - **National Centre for Vector Borne Disease Control (NCVBDC)** – malaria, dengue, chikungunya stats
  - **Ministry of Health & Family Welfare (MoHFW)** – national and state-level health reports
  - **Indian Meteorological Department (IMD)** – temperature, rainfall, humidity reports
  - **World Health Organization (WHO)** – climate and disease guidelines
  - **State Health Department Portals**
  - **Research articles and climate-health studies**
  - **Government open data portals**
- ✓ **Data Used**
  - State-wise annual disease cases
  - Temperature variations
  - Rainfall data
  - Humidity levels
  - Extreme weather events (floods, droughts, cyclones)

3. Variables Used in the Study:

- ✓ **Dependent Variables (Disease Data)**
  - Malaria, Dengue, Chikungunya
  - Diarrhoea, Cholera, Typhoid
  - Heat stroke, Dehydration

- Leptospirosis
- Respiratory diseases

✓ **Independent Variables (Climate Factors)**

- Temperature increase
- Rainfall patterns
- Humidity levels
- Cyclones, floods, droughts
- Seasonal variations

✓ **Demographic Variables**

- State population
- Density
- Urban vs rural distribution

4. <u>Data Preparation and Cleaning:</u> To ensure accuracy, the collected data is cleaned before analysis.

✓ **Data Cleaning Steps**

- Removing duplicates
- Handling missing values
- Correcting inconsistent entries
- Formatting dates and numeric values
- Merging datasets from different sources

✓ **Data Transformation**

- Converting raw files into CSV/Excel format
- Creating new columns (e.g., disease rate per 1 lakh population)
- Standardizing units (°C, mm, percentages)
- Categorizing states into zones (North, South, East, West, North-east)

✓ **Data Integration-** Combined datasets for:

- Climate data
- Disease data
- State-level demographics

5. <u>Tools and Software Used:</u>

✓ **Microsoft Power BI-** The main tool used for:

- Data cleaning
- Data modelling
- Visualization
- Dashboard creation

✓ **Microsoft Excel-** Used for:

- Preliminary cleaning
- Filtering
- Arranging datasets

6. <u>Power BI Analysis Process:</u>

✓ **Data Import-** Import cleaned Excel/CSV files into Power BI

✓ **Data Modelling**

- Setting relationships between climate and disease tables
- Creating calculated columns & measures
- Normalizing data based on population

✓ **Visualisation-** Power BI visuals used include:

- Line charts (trend over years)
- Bar charts (state-wise comparison)
- Heat maps (disease hotspots)
- Pie charts (disease proportion)
- Scatter plots (correlation analysis)
- Filled maps (geographical display)

✓ **Dashboard Creation**

- Combining visuals into interactive dashboards
- Adding slicers for state-wise, year-wise, and disease-wise filtering
- Designing a user-friendly layout for interpretation

7. Data Analysis Techniques:

✓ **Trend Analysis-** Identifies how disease numbers change over time

✓ **Correlation Analysis-** Shows relationships between temperature, rainfall, and diseases

✓ **Comparative Analysis-** Compares states to identify high-risk regions

✓ **Seasonal Analysis-** Determines how monsoon, summer, and winter impact disease peaks

✓ **Geographical Analysis-** Uses Power BI maps to show disease spread regionally

8. Interpretation of Results:

- Results are interpreted based on patterns displayed in visuals
- States with high vulnerability are identified
- Climate variables are linked to disease behaviour
- Recommendations are developed for policymakers

9. Ethical Considerations:

- Only publicly available and government-approved data is used
- No personal or private patient data is included
- Data is used strictly for educational and research purposes

10. Summary of Methodology: The study follows a structured approach involving:

1. Collection of climate and disease data
2. Cleaning and organizing the dataset
3. Importing into Power BI
4. Data modelling and visualization
5. Interpretation of insights
6. Identification of climate–disease patterns

SIGNIFICANCE OF THE STUDY

Global warming has emerged as one of the most serious threats to human health in the 21st century. India, being one of the most climatically diverse and densely populated countries, faces increasing risks from climate-induced diseases. This study carries major significance because it not only maps these risks across different Indian states but also uses Power BI to transform raw climate and health data into meaningful insights. The findings contribute to public health planning, climate policy development, and academic advancement.

1. Importance for Public Health and Disease Prevention

The study is significant in highlighting the strong connection between rising temperatures, unpredictable rainfall, and the increasing burden of diseases such as:

- Dengue
- Malaria
- Cholera
- Heatstroke
- Respiratory illnesses
- Waterborne infections

By identifying state-wise risk zones and disease patterns, the study helps public health departments:

- Prepare early warnings
- Implement preventive measures
- Allocate resources efficiently
- Strengthen seasonal health campaigns

This ensures timely action and reduces disease outbreaks.

2. Contribution to Climate Change Awareness in India

India is among the top countries exposed to climate vulnerability. This project contributes to national awareness by showing:

- How warming trends directly affect human health
- How vector-borne diseases spread faster with humidity
- How floods and heavy rainfall increase waterborne diseases
- How extreme heat events lead to hospitalization and mortality

The study reinforces the need for climate-resilient infrastructure and long-term environmental planning.

3. Enhancing Decision-Making Through Data Analytics

The use of Power BI adds a modern and highly practical dimension to the study. The significance lies in:

- Converting complex data into simple dashboards
- Allowing decision-makers to compare states interactively
- Helping health administrators visualize hotspots
- Supporting predictive forecasting of future disease trends

This shows how data analytics can transform healthcare and environmental management.

4. Importance for Government and Policymakers

The findings of this study can support central and state governments in:

- Drafting heatwave action plans
- Designing climate-health policies
- Strengthening vector-borne disease control programs
- Evaluating climate adaptation strategies
- Planning disaster management activities

The project provides evidence-based insights necessary for policymaking at both state and national levels.

5. Contribution to Academic and Research Communities

This study serves as an important academic resource because it:

- Combines climate science, health studies, and data analytics
- Demonstrates how interdisciplinary research can address real-world problems
- Provides a strong case study for BBA/MBA/B.Sc. researchers
- Offers a model for students to execute analytical projects

It strengthens research skills and promotes the use of analytics tools in academic work.

6. Supporting NGOs, Hospitals, and Health Agencies

The insights from the project can help:

- NGOs create targeted awareness programs
- Hospitals prepare for seasonal patient load
- Health departments increase surveillance in high-risk areas
- Environmental groups promote sustainable practices

It helps these organizations develop practical, community-focused interventions.

7. Forecasting Future Climate and Disease Risks

A major significance of this study is its predictive capability. Through Power BI forecasting and trend analysis, the project can:

- Predict rising disease levels
- Identify upcoming climate-induced risks
- Highlight areas where immediate action is needed
- Guide long-term climate-health strategies

This makes the study valuable for future planning and risk mitigation.

8. Empowering Students With Real-World Skill Development

The project showcases the importance of analytical tools like Power BI.
With guidance and training from **Ardent Computech Pvt. Ltd.**, students gain:

- Hands-on experience in data analytics
- Ability to clean, model, and visualize data
- Skills needed in today's data-driven industries
- Real-world project experience for their portfolio

This increases career readiness and strengthens technical competence.

Overall, the significance of this study lies in its ability to connect climate change with real public health challenges and present them through a powerful analytical framework.

It benefits governments, health agencies, students, researchers, and communities by offering clear, data-driven insights into the urgent relationship between global warming and disease patterns in India.

IMPLEMENTATION OF THE PROJECT

The implementation phase of this project involved the systematic use of climate data, disease records, and Power BI tools to analyze the impact of global warming on disease patterns across different states of India. The process included data collection, data cleaning, model development, visualization, and interpretive analysis. Each step was executed in a structured manner to ensure accuracy, reliability, and meaningful insights.

1. Data Collection and Compilation

The first step in implementation was gathering relevant datasets from credible and publicly accessible sources. Climate variables such as temperature, rainfall, humidity, heatwaves, and seasonal variations were collected state-wise from sources like:

- Indian Meteorological Department (IMD)
- Ministry of Health & Family Welfare (MoHFW)
- National Centre for Vector-Borne Diseases
- WHO and IPCC reports (supporting data)

Disease-related datasets included:

- Vector-borne diseases (Dengue, Malaria, Chikungunya)
- Waterborne diseases (Cholera, Diarrhoea)
- Heat-related illnesses (Heatstroke, Dehydration)
- Respiratory diseases (Asthma, Bronchitis due to heat & pollution)

All datasets were compiled into Excel files to maintain consistency and ease of import into Power BI.

2. Data Cleaning and Pre-Processing

Raw data often contains missing values, inconsistencies, or variations in formats. Before analysis, the datasets were cleaned using:

- **Handling Missing Values**
  - Replacing missing temperature or rainfall readings using monthly averages
  - Removing duplicate entries
  - Fill-down method for incomplete state records
- **Standardization**
  - Standardizing units (°C for temperature, mm for rainfall, % for humidity)
  - Ensuring common year ranges across datasets
- **Categorization:** States were categorized based on climate zones:

- Tropical
- Dry
- Humid Subtropical
- Mountainous

This made the analysis more structured and comparable.

3. Importing Data into Power BI

After cleaning, the datasets were imported into Power BI using:

- **Power Query Editor** for transformations
- **Data modeling tools** to establish relationships between climate variables and disease incidence

All climate and health tables were linked through fields such as:

- State Name
- Year
- Disease Type

This established a unified analytical model.

4. Data Modeling and Relationship Building

Using the **Model View** in Power BI:

- **Creation of Star Schema**
  - **Fact Table:** Disease Count
  - **Dimension Tables:** State, Climate Variables, Year

5. Visualization and Dashboard Development

This was the core practical part of the project where meaningful insights were presented through interactive dashboards. Multiple visual tools were used, including:

- **Map Visuals**
- Choropleth maps for state-wise disease distribution
- Heat intensity maps for climate vulnerability
- **Time-Series Charts**
- Temperature trend lines
- Disease incidence over years
- Rainfall–disease correlation charts

- **Comparative Analysis Visuals**

- Clustered column charts
- Stacked bar charts
- Bubble charts

These visuals enabled easy comparison across states and climates.

6. Correlation and Trend Analysis

Once the dashboards were built, advanced Power BI analytical tools were used to identify trends:

- **Correlation Patterns**

- High temperature → Increased dengue outbreaks
- Excess rainfall → Rise in cholera and diarrhoea
- Humidity → Rapid mosquito breeding
- Heatwaves → Spike in heatstroke cases

- **Forecasting Tools:** Power BI's forecasting feature was used to project:

- Future disease risks
- Temperature rise trends
- Expected vulnerability levels

This improved predictive accuracy of the study.

7. Validation and Cross-Checking

To ensure the accuracy of results, the findings were validated using multiple techniques:

- Cross-checking with government health reports
- Verifying climate change patterns using IPCC reports
- Checking the reliability of disease peaks with historical data

Where discrepancies emerged, data was corrected or adjusted logically.

8. Documentation and Report Preparation

Finally, all insights, Power BI dashboards, and interpretations were documented to create:

- A professional written report
- A well-structured documentary/visual narrative
- Stakeholder-friendly presentations

Special support and technical training from **Ardent Computech Pvt. Ltd.** helped in shaping the technical aspects of Power BI visualization, dashboard creation, and model building.

ANALYSIS

- **DASHBOARD 1: National Climate–Disease Overview: (FIG No. 1)**

Cards (Top Summary Indicators):

- **Sum of Cases Reported:**
  Shows the total number of climate-related disease cases reported across all selected Indian states.
  This gives a national-level snapshot of disease burden caused by climate change.
- **Sum of Population:**
  Represents the total population covered in the dataset, helping to understand exposure level.

Sum of Cases Reported by Disease Name(column chart):

- Displays disease-wise comparison of total cases.
- Diseases included: **Malaria, Respiratory Illness, Cholera, Dengue, Heat Stroke**.
- **Insight:**
  Vector-borne and respiratory diseases dominate, proving strong linkage with temperature rise, humidity, and air quality.

Sum of Cases Reported by Disease Type (Donut Chart):

- Breaks cases into **Disease Categories**:
  - Vector-Borne (largest share ~47%)
  - Air-Quality related
  - Water-Borne
  - Climate-Heat
- **Insight:**
  Vector-borne diseases are the most climate-sensitive, driven by humidity and rainfall.

Sum of Cases Reported by Air Quality Index (line Chart)

- Compares disease cases against AQI levels (100–300).
- **Insight:**
  Higher AQI values are linked with higher disease cases, especially respiratory illnesses.
- Proves **climate change + pollution = health crisis**.

Age Group & State Slicers:

- Allow interactive filtering by:
  - Age groups (0–14, 15–30, 31–50, 51+)
  - Disease name
  - State
- **Purpose:**
  Enables customized analysis for policymakers and health planners.

- **DASHBOARD 2: State-Wise Disease Impact Analysis: (FIG No. 2)**

Map Visual – Cases Reported by State

- Displays geographic distribution of disease burden.
- Darker regions indicate higher cases.
- **Insight:**
  States like **West Bengal, Karnataka, Uttar Pradesh, Gujarat** show higher vulnerability.

Bar Chart – Sum of Cases Reported by State

- Ranks states based on total disease cases.
- **Insight:**
  Highlights climate-vulnerable states requiring urgent health intervention.

Stacked Bar – Disease Name by State

- Shows which diseases dominate in which states.
- **Insight:**
  - Dengue & malaria dominate humid states
  - Heat stroke appears more in hot, dry states

Tabular Climate–Health Matrix Columns include:

- Rainfall
- Humidity
- Air Quality Index
- Healthcare Infrastructure Score
- **Insight:**
  Shows how **poor infrastructure + adverse climate = higher disease impact**.

- **DASHBOARD 3: Climate Factors vs Disease Patterns: (FIG No. 3)**

Scatter Plot – Rainfall vs Cases (with Humidity)

- Each point represents a disease occurrence.
- **Insight:**
  High rainfall and humidity strongly correlate with increased dengue, malaria, and cholera.

Line Chart – Dengue, Malaria & Diarrhoeal Cases Over Time

- Month-wise disease trend (2023).
- **Insight:**
  Sharp rise during monsoon months confirms climate dependency.

Bar Chart – Heatstroke Deaths vs Respiratory Hospitalization

- Shows interaction between extreme heat and respiratory stress.
- **Insight:**
  Heatwaves worsen respiratory conditions, increasing mortality risk.

Monthly Hospitalisation Table

- Shows seasonal spike in hospital admissions.
- **Insight:**
  July–August shows peak burden due to monsoon and heat stress.

- **DASHBOARD 4: Seasonal Disease Trends: (FIG No. 4)**

Line Chart – Max Temperature by Month

- Shows rising temperature trend from Jan to summer months.
- **Insight:**
  Rising temperature precedes heatstroke and respiratory cases.

Cluster column Chart – Total Rainfall by Month

- Shows monsoon peaks.
- **Insight:**
  Rainfall spikes align with vector- and water-borne disease outbreaks.

Table – Extreme Rain Events & PM2.5

- Combines climate extremes with pollution.
- **Insight:**
  Climate events amplify pollution-related health risks.

Disease Type Donut Chart

- Reinforces national disease composition during seasonal variation.

- **DASHBOARD 5: High-Risk Zones Identification: (FIG No. 5)**

Bar Chart – Deaths Reported by State

- Identifies states with highest climate-linked mortality.
- **Insight:**
  States like **Maharashtra, Kerala, Rajasthan** show high death counts.

Table – Incidence Rate, Population & Deaths

- Helps normalize disease impact per population.
- **Insight:**
  Even lower-population states can show high risk if incidence rate is high.

Map – High-Risk Mortality Zones

- Visual identification of critical states.
- **Purpose:**
  Useful for disaster preparedness and policy planning.

- **DASHBOARD 6: Disease Type Deep Dive: (FIG No. 6)**

Treemap – Cases by Disease Type

- Confirms dominance of vector-borne diseases.

Bar Chart – Deaths by Disease Name

- **Insight:**
  Respiratory illnesses cause the highest deaths, proving the danger of air quality deterioration.

Healthcare Infrastructure, Urban/Rural & Poverty Indicators

- Shows socio-economic influence on health outcomes.
- **Insight:**
  Poorer regions suffer higher mortality due to limited healthcare access.

Slicer: name of the states

- It shows the name of the states

➢ **DASHBOARD 7: District-Level & Rural–Urban Comparison: (FIG No. 7)**

Cards – Rural vs Urban

- Rural areas show:
    - Higher cases
    - Higher deaths
- **Insight:**
  Climate impact is more severe in rural India due to weak infrastructure.

line Chart – Cases by Age Group

- Highest cases in **31–50 and 51+** age groups.
- **Insight:**
  Working population is highly exposed.

Map – Urban vs Rural Distribution

- Shows spatial inequality in disease burden.

✓ **OVERALL CONCLUSION FROM VISUALS**

Your dashboards clearly prove that:

- Global warming directly affects disease patterns
- Climate variables (heat, rainfall, humidity, pollution) drive disease outbreaks
- Vector-borne and respiratory diseases are most climate-sensitive
- Rural and high-population states face greater risk
- Power BI effectively converts complex data into actionable insights

LIMITATIONS OF THE PROJECT

Despite providing meaningful insights into the impact of global warming on disease patterns across different states of India, this project has certain limitations that should be acknowledged for clarity and academic completeness.

1. **Dependence on Secondary Data**:The study is entirely based on secondary data collected from government reports, public datasets, and published sources. Any inaccuracies, delays, or underreporting in these sources may affect the results of the analysis.
2. **Limited Time Period Coverage**: The analysis covers a limited time span due to data availability. Climate change is a long-term phenomenon, and longer historical datasets could provide more accurate trend and impact analysis.
3. **Aggregated State-Level Data**: Most of the analysis is conducted at the state level. District-level or city-level data could reveal more precise local impacts but were not fully available for all regions.
4. **Restricted Disease Categories**: The project focuses only on selected climate-sensitive diseases such as vector-borne, waterborne, respiratory, and heat-related illnesses. Other indirect health impacts like mental health issues or nutritional disorders are not included.
5. **Correlation-Based Analysis**: The study identifies correlations between climate variables and disease patterns but does not establish direct causation. Medical, genetic, and socio-behavioral factors were beyond the scope of this project.
6. **Data Quality and Missing Values**: Some datasets contained missing or inconsistent values, which were managed using standard cleaning techniques. However, this may slightly influence precision.
7. **Limited Healthcare Infrastructure Indicators**: Healthcare capacity is measured using selected indicators only. More detailed variables such as hospital bed availability, doctor-to-patient ratio, and emergency response time were not included.
8. **Tool-Based Constraints**: The analysis is conducted using Power BI, which focuses primarily on visualization and descriptive analytics. Advanced statistical modeling and machine learning techniques were not applied.
9. **External Factors Not Considered:** Factors such as migration, lifestyle changes, vaccination coverage, and policy interventions were not incorporated into the analysis.

- **Overall:** Although the project has limitations, it successfully provides a **strong data-driven overview of climate-induced disease patterns in India**. The findings remain valuable for academic study, policy awareness, and future research expansion.

FINDINGS OF THE STUDY

Based on the analysis of climate and health data across different states of India using Power BI, the following key findings have been observed. These findings clearly demonstrate the strong relationship between global warming, climate variability, and the increasing burden of diseases in India.

1. Rise in Climate-Induced Disease Burden

The analysis shows a high overall disease burden, with more than **1.26 lakh cases** reported across the selected states. This confirms that climate change is significantly influencing public health outcomes in India.

2. Vector-Borne Diseases Are the Most Affected

Among all disease categories, **vector-borne diseases account for the highest share (approximately 47%)** of total reported cases. Dengue and malaria show sharp increases in states with high humidity and heavy rainfall, proving strong climate sensitivity.

3. Respiratory Illnesses Cause the Highest Mortality

Although vector-borne diseases have the highest number of cases, **respiratory illnesses contribute to the highest number of deaths**. Poor air quality combined with rising temperatures significantly increases respiratory hospitalizations and fatalities.

4. Strong Relationship Between Climate Factors and Disease Patterns

The analysis reveals clear correlations:

- Increased rainfall and humidity lead to higher dengue, malaria, and cholera cases.
- Extreme temperatures and heatwaves result in increased heatstroke cases and deaths.
- Higher Air Quality Index (AQI) values are associated with increased respiratory illnesses.

These findings confirm that climate variables play a direct role in disease outbreaks.

5. Significant State-Wise Variation in Disease Impact

The study highlights wide variation across states:

- States such as **West Bengal, Karnataka, Maharashtra, Uttar Pradesh, and Gujarat** report higher disease incidence.
- Coastal and monsoon-affected states show greater vulnerability to vector- and water-borne diseases.
- Heat-prone states report more heat-related illnesses.

This indicates the need for region-specific health and climate strategies.

6. Seasonal Peaks in Disease Occurrence

Seasonal trend analysis shows:

- Disease cases rise sharply during **monsoon months (July–September)**.
- Heatstroke cases increase during peak summer months.
- Respiratory cases rise during periods of high temperature and pollution.

Seasonality plays a crucial role in disease spread.

7. Rural Areas Are More Vulnerable Than Urban Areas

The rural–urban comparison reveals:

- Higher number of cases and deaths in rural areas.
- Limited healthcare infrastructure and delayed access contribute to higher mortality.
  This highlights the unequal impact of climate change on vulnerable populations.

8. Working-Age Population Is Most Affected

Age-group analysis shows that individuals aged **15–30 and 31–50 years** report the highest number of cases. This suggests increased exposure due to outdoor work, travel, and occupational risk.

9. Healthcare Infrastructure Influences Mortality Outcomes

States with lower healthcare infrastructure scores report higher death rates, even when case numbers are similar. This highlights the importance of strong healthcare systems in reducing climate-related health impacts.

10. Effectiveness of Power BI in Health–Climate Analysis

The use of Power BI enabled:

- Clear visualization of complex climate–health relationships.
- Identification of high-risk zones and disease hotspots.
- Interactive analysis for better decision-making.

This confirms the effectiveness of data analytics tools in addressing real-world environmental and health challenges.

Overall Finding

The study conclusively finds that **global warming significantly influences disease patterns in India**, with climate variables such as temperature, rainfall, humidity, and air quality acting as key drivers. Without timely intervention, climate-induced health risks are expected to increase further.

SUGGESTIONS / RECOMMENDATIONS

Based on the findings of the study on climate-induced diseases across different states of India, the following suggestions are proposed to reduce health risks, strengthen preparedness, and support sustainable decision-making.

1. Strengthening Climate-Based Early Warning Systems: Government and health authorities should develop climate-linked disease early warning systems. By monitoring temperature, rainfall, humidity, and air quality in real time, potential outbreaks of dengue, malaria, cholera, and heat-related illnesses can be predicted and controlled in advance.

2. State-Specific Health and Climate Action Plans: Since disease impact varies widely across states, uniform policies may not be effective. Each state should design climate-health action plans based on:

- Local climate conditions
- Disease vulnerability
- Seasonal patterns

This will improve targeted interventions and efficient resource allocation.

3. Improving Vector Control and Sanitation Measures: To reduce vector-borne and water-borne diseases:

- Regular mosquito control programs should be strengthened
- Proper drainage systems should be maintained
- Clean drinking water access should be ensured, especially during monsoons

Community participation should be encouraged for sustainable results.

4. Heatwave Preparedness and Public Awareness: Heat-prone states should implement:

- Heatwave action plans
- Public cooling centres
- Awareness campaigns on hydration and heat safety
- Flexible work hours during extreme heat

These steps can significantly reduce heatstroke cases and deaths.

5. Air Quality Management and Pollution Control: Given the strong link between poor air quality and respiratory diseases:

- Strict emission control policies should be enforced
- Promotion of green transport and renewable energy
- Expansion of urban green spaces

These measures will help reduce pollution-related health risks.

6. Strengthening Rural Healthcare Infrastructure: Rural areas show higher vulnerability due to limited healthcare access. Therefore:

- Primary health centers should be upgraded
- Mobile medical units should be introduced
- Emergency response systems should be strengthened

This will help reduce climate-related mortality in rural regions.

7. Use of Data Analytics Tools for Policy Making: Government departments, hospitals, and NGOs should adopt tools like **Power BI** for:

- Real-time disease monitoring
- Risk zone identification
- Seasonal forecasting
- Evidence-based decision making

This project demonstrates the practical value of data-driven governance.

8. Integration of Climate and Health Policies: Climate change and public health policies should not be treated separately. Integrated planning is required to:

- Build climate-resilient healthcare systems
- Reduce long-term disease burden
- Improve disaster preparedness

9. Encouraging Academic and Research Collaboration: Universities, research institutions, and training organizations like **Ardent Computech Pvt. Ltd.** should promote interdisciplinary research combining climate science, health, and data analytics to address emerging challenges.

10. Continuous Data Collection and Monitoring: Regular updating of climate and health datasets is essential. Continuous monitoring will help:

- Track changing disease patterns
- Improve forecast accuracy
- Support long-term planning

Overall Recommendation

To effectively combat climate-induced diseases, India must adopt a **data-driven, preventive, and state-specific approach**, supported by strong healthcare systems, public awareness, and modern analytical tools.

OVERALL CONCLUSION

This project concludes that **global warming has a significant and measurable impact on disease patterns across different states of India**. Rising temperatures, changing rainfall patterns, increased humidity, and deteriorating air quality are no longer only environmental concerns but have emerged as major public health challenges. The analysis clearly demonstrates that climate change is directly and indirectly responsible for the increasing burden of climate-sensitive diseases in the country.

The Power BI–based analysis reveals that **vector-borne diseases** such as dengue and malaria account for the highest proportion of cases, particularly in states with high humidity and heavy rainfall. **Respiratory illnesses**, strongly linked to air pollution and rising temperatures, contribute to the highest number of deaths, highlighting the serious health consequences of worsening air quality. **Heat-related illnesses**, including heatstroke, show a clear rise during extreme temperature periods, especially in heat-prone regions.

State-wise and seasonal analysis confirms that the impact of climate change on health is **not uniform across India**. Coastal, monsoon-affected, and densely populated states experience a higher disease burden, while rural areas are more vulnerable due to limited healthcare infrastructure and delayed medical access. The working-age population is found to be the most affected, indicating increased exposure to climate risks through occupational and environmental factors.

A major strength of this study lies in the **use of Power BI as a data analytics tool**, which effectively transformed complex climate and health data into interactive dashboards and visual insights. The use of data visualization enabled clear identification of high-risk zones, disease trends, and climate–health relationships, making the analysis useful for policymakers, healthcare administrators, researchers, and academic purposes.

Overall, the project establishes that addressing climate-induced health risks requires a **data-driven, preventive, and region-specific approach**. Strengthening healthcare infrastructure, improving climate preparedness, controlling pollution, and adopting modern analytical tools are essential steps toward reducing the future health burden of climate change. The study also demonstrates how interdisciplinary approaches—combining climate science, public health, and data analytics—can contribute to sustainable decision-making and effective public health planning.

In conclusion, this project not only fulfills its academic objectives but also provides practical insights that can support climate-resilient healthcare strategies in India. With continuous data monitoring and informed policy action, the adverse health impacts of global warming can be mitigated effectively.

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ANEXTURE

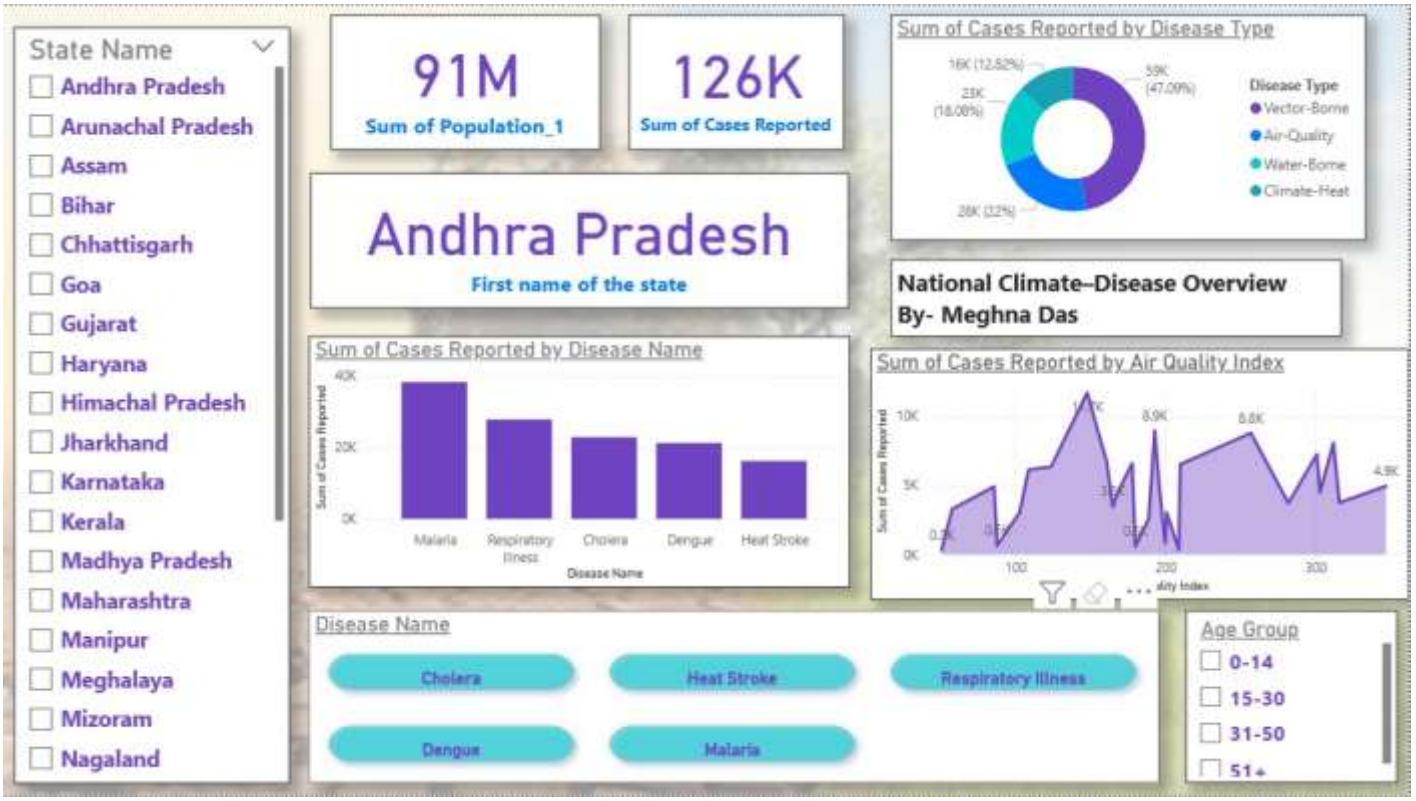


FIG No. 1

Sum of Rainfall	name of the state	Disease Name	Sum of Humidity (%)	Sum of Air Quality Index	Sum of Healthcare Infra Score (NFHS-5: % HHs Covered by Health Scheme)
671.20	Maharashtra	Dengue	55.90	189	84.60
661.00	Haryana	Malaria	35.30	301	82.80
1,197.60	Andhra Pradesh	Cholera	52.40	88	82.20
1,053.40	Gujarat	Malaria	68.50	312	79.90
1,557.50	Jharkhand	Respiratory Illness	45.70	161	78.50
883.60	Madhya Pradesh	Respiratory Illness	93.90	303	78.40
37,483.50			1,802.40	5284	1,755.40

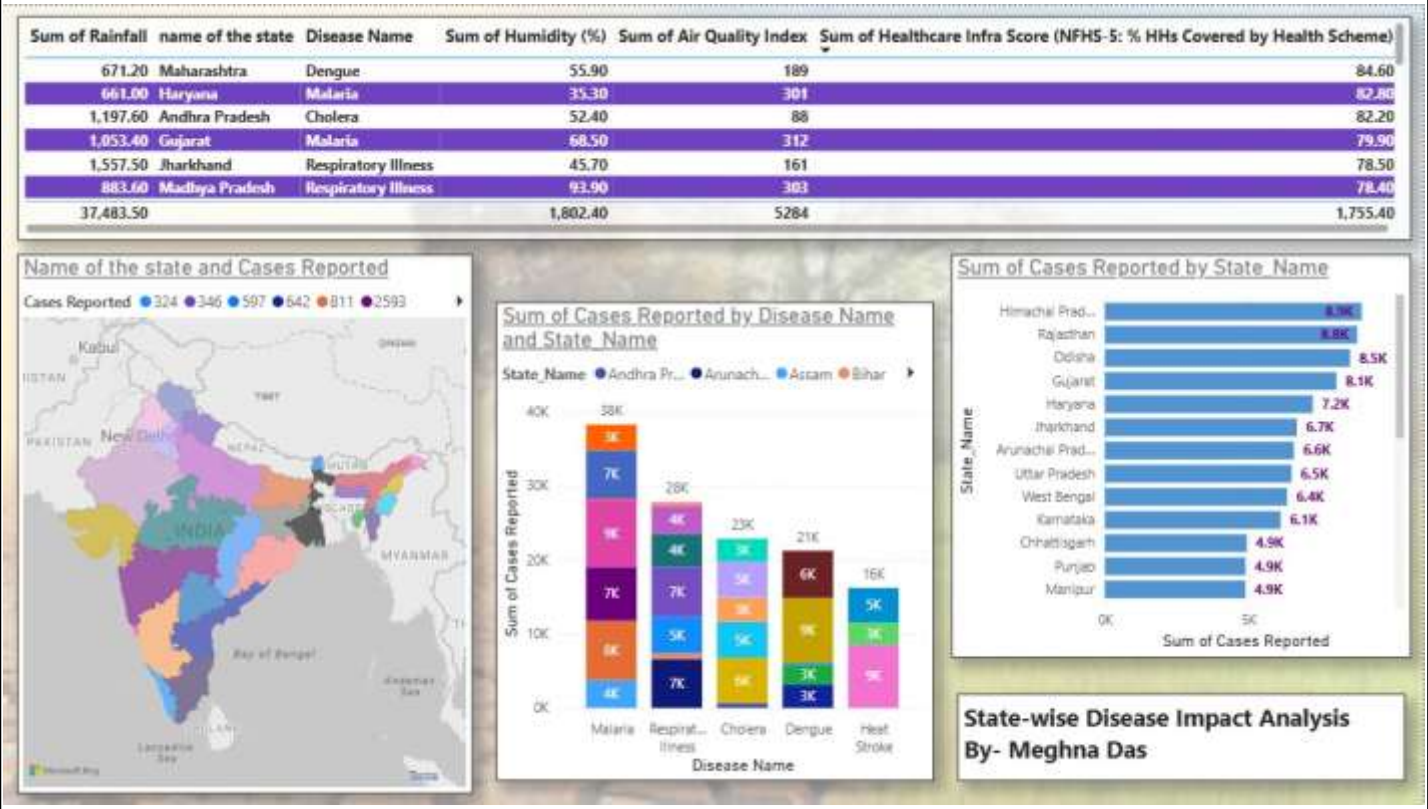


FIG No. 2

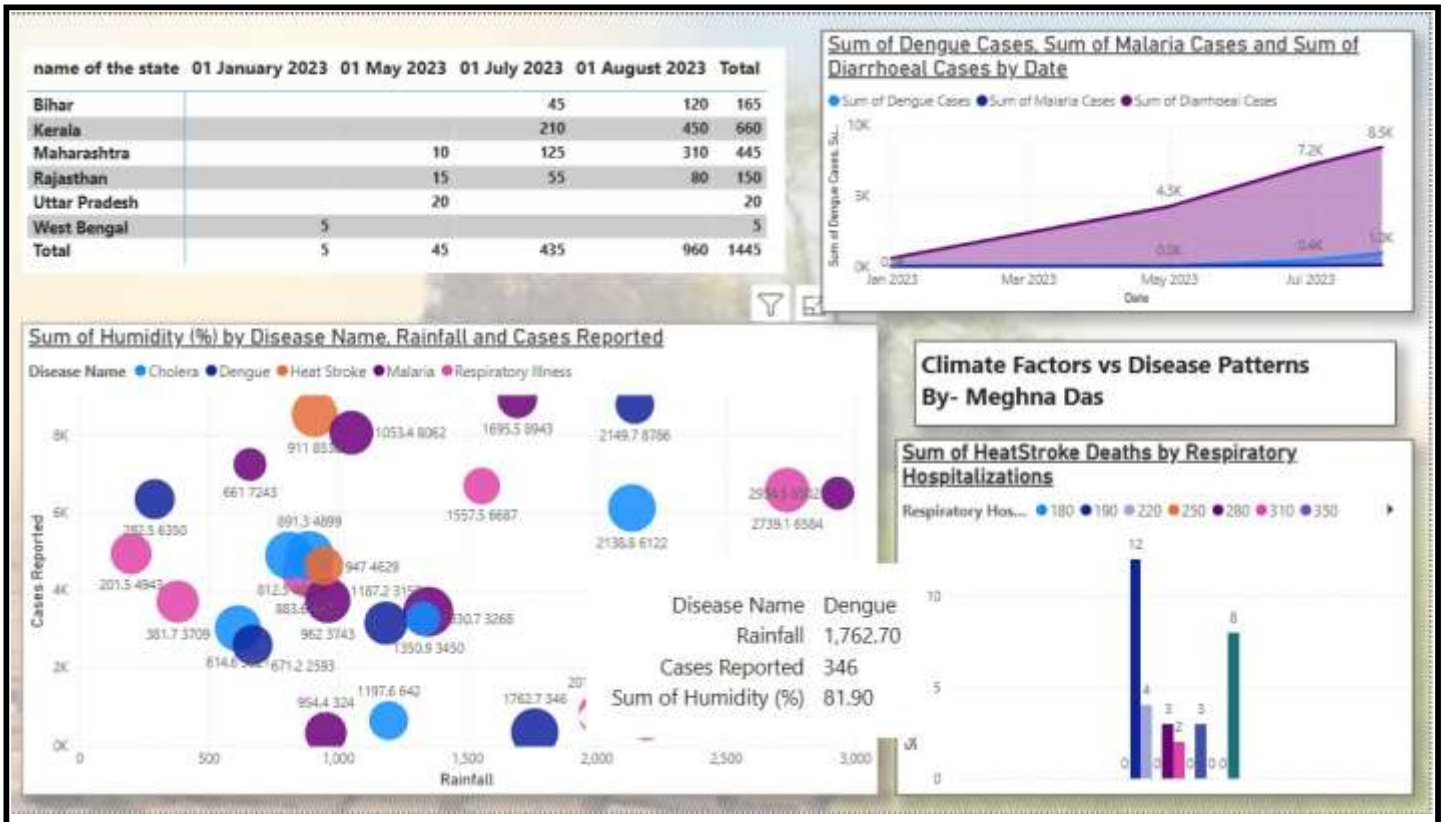


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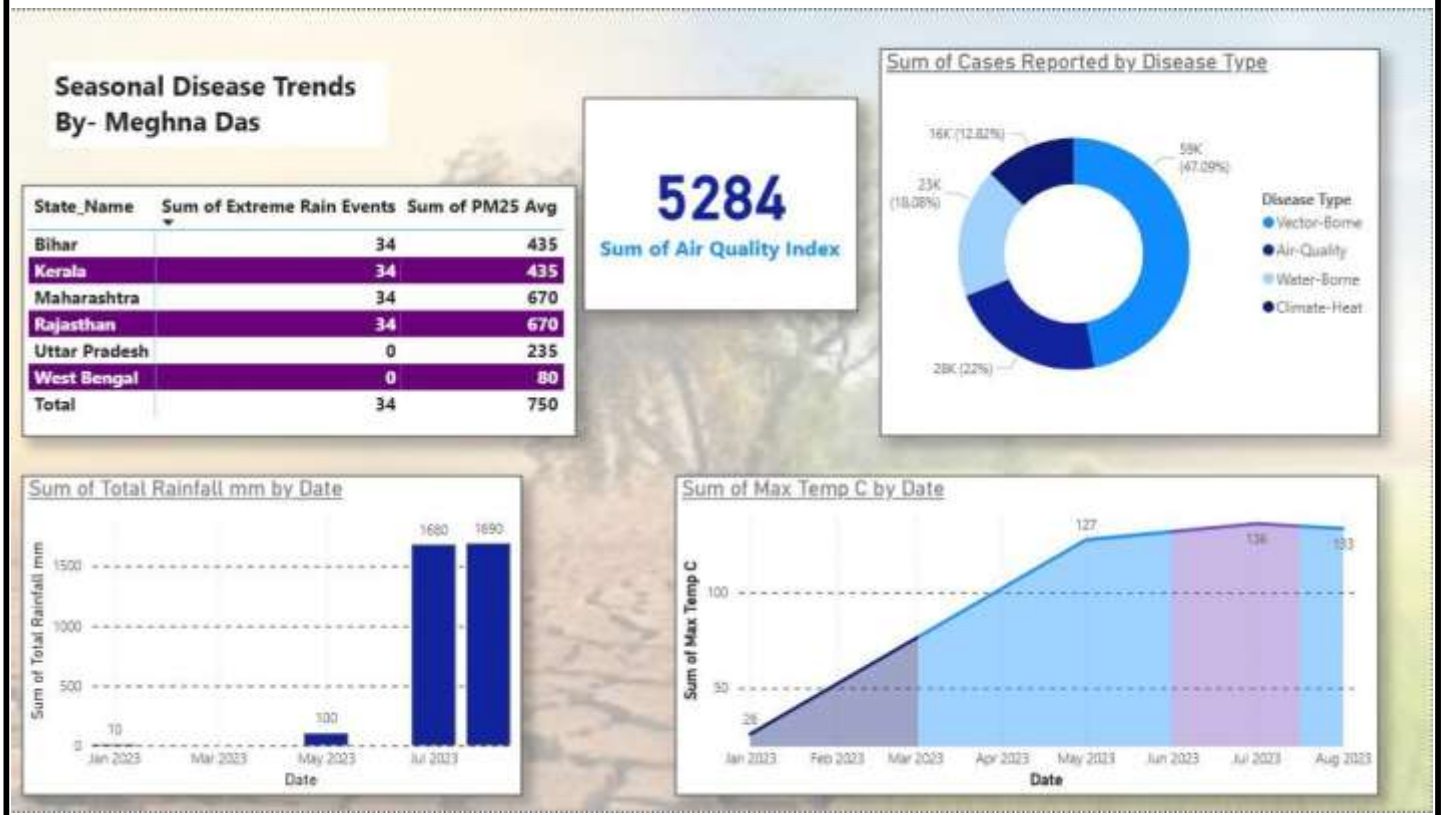
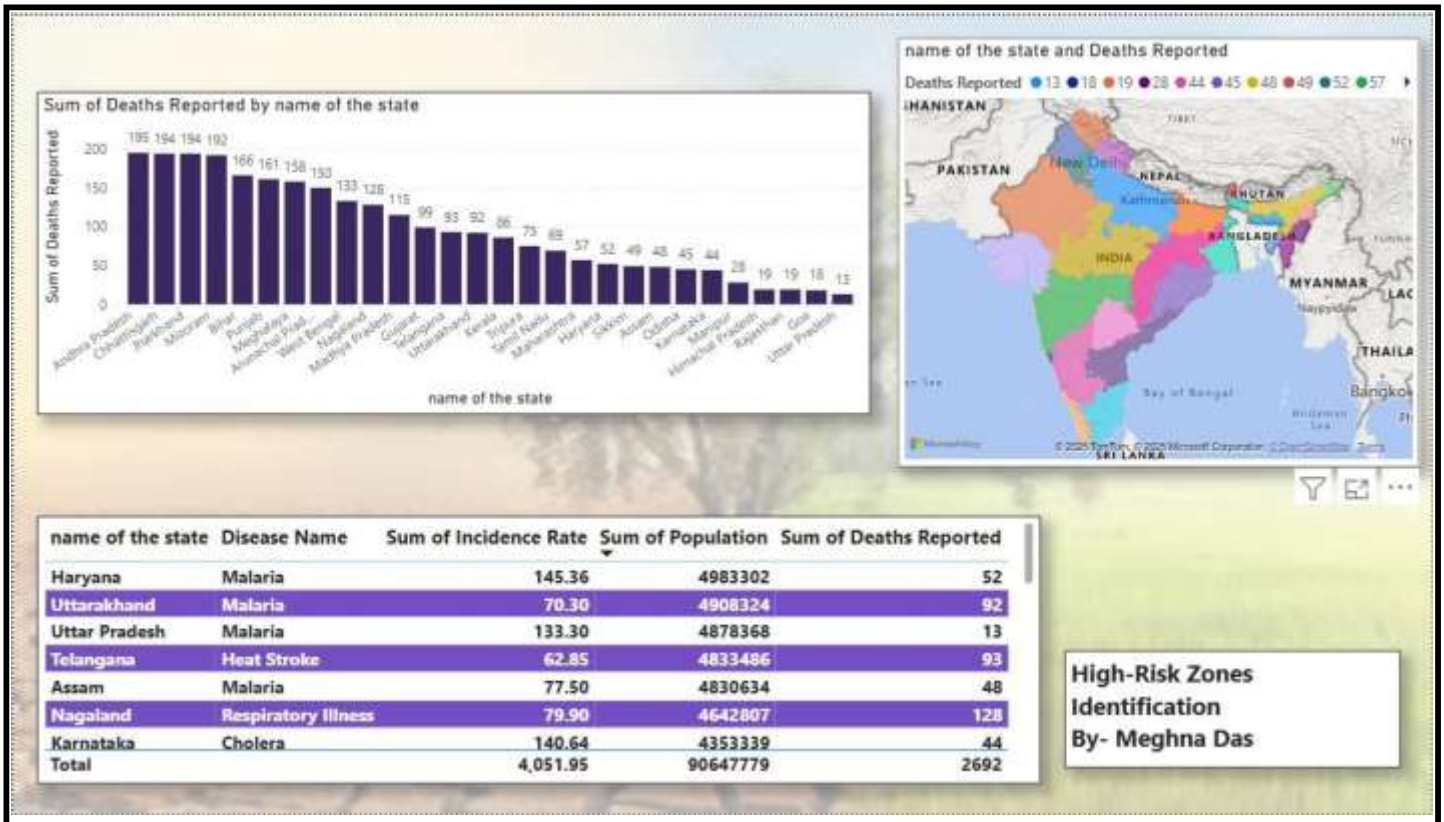


FIG No. 4



name of the state	Disease Name	Sum of Incidence Rate	Sum of Population	Sum of Deaths Reported
Haryana	Malaria	145.36	4983302	52
Uttarakhand	Malaria	70.30	4908324	92
Uttar Pradesh	Malaria	133.30	4878368	13
Telangana	Heat Stroke	62.85	4833486	93
Assam	Malaria	77.50	4830634	48
Nagaland	Respiratory Illness	79.90	4642807	128
Karnataka	Cholera	140.64	4353339	44
Total		4,051.95	90647779	2692

FIG No. 5

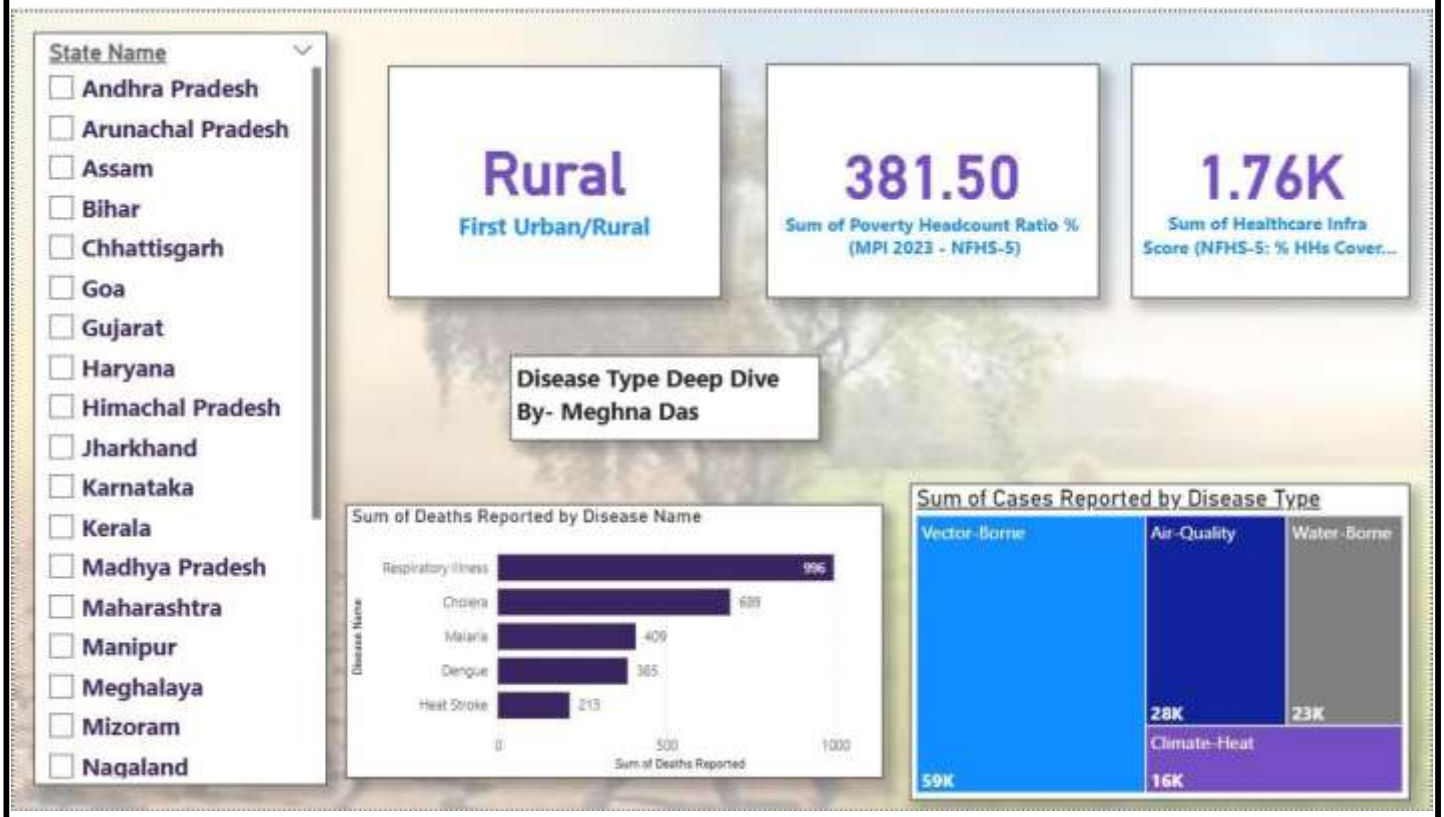


FIG No. 6

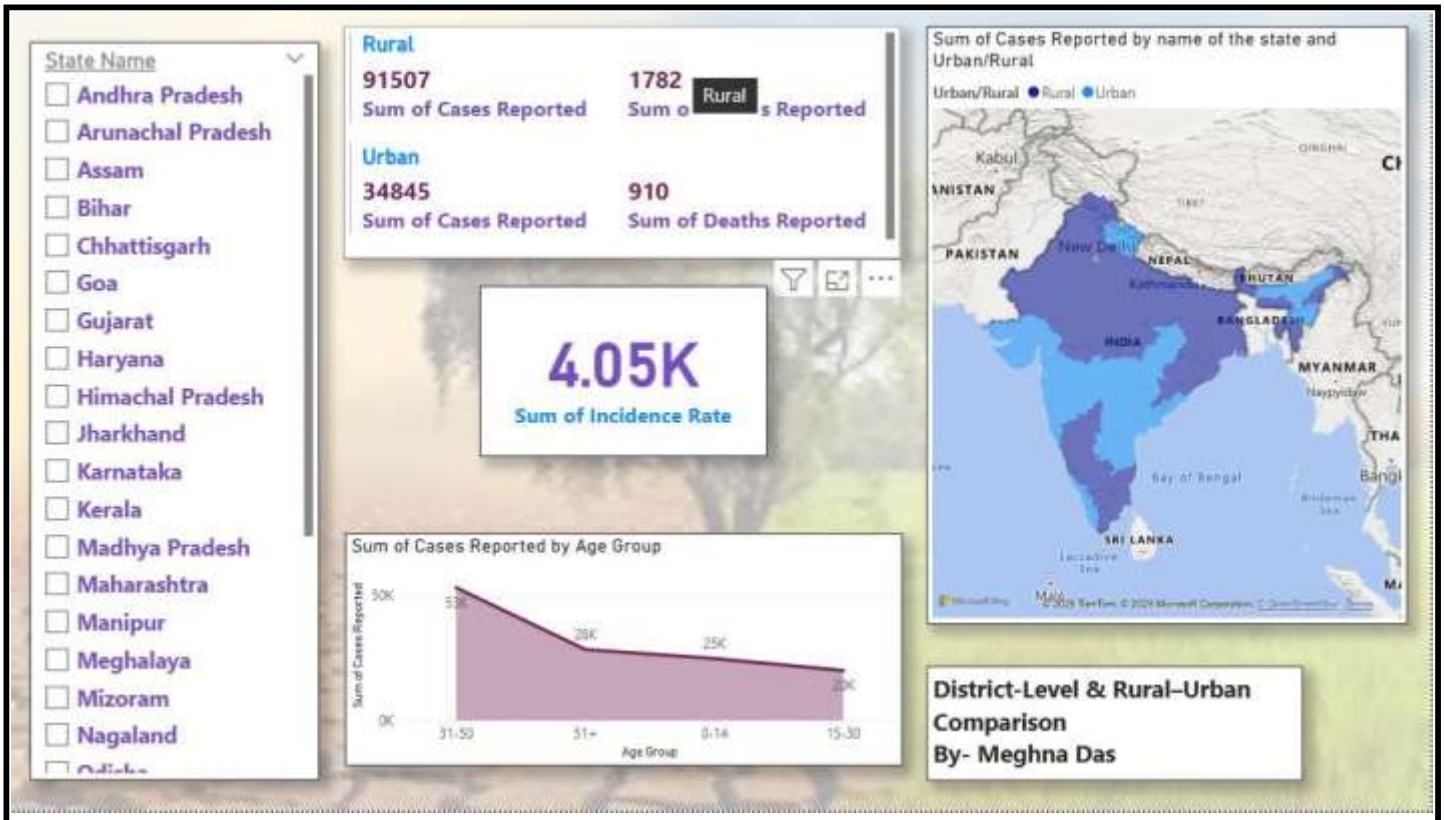

State Name
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Chhattisgarh
Goa
Gujarat
Haryana
Himachal Pradesh
Jharkhand
Karnataka
Kerala
Madhya Pradesh
Maharashtra
Manipur
Meghalaya
Mizoram
Nagaland
Rural
91507
Sum of Cases Reported
1782
Rural
Urban
34845
Sum of Cases Reported
910
Sum of Deaths Reported
4.05K
Sum of Incidence Rate
Sum of Cases Reported by Age Group
Sum of Cases Reported
Age Group
31-50
51+
0-14
15-30
Sum of Cases Reported by name of the state and Urban/Rural
Urban/Rural
Rural
Urban
Kabul
PAKISTAN
New Delhi
NEPAL
BHUTAN
INDIA
MYANMAR
SRI LANKA
Bay of Bengal
District-Level & Rural–Urban Comparison
By- Meghna Das

FIG No. 7